

6. Trigonometry		
6.1	Knowledge and use of $\tan \theta = \frac{\sin \theta}{\cos \theta}$, and $\sin^2 \theta + \cos^2 \theta = 1$.	
6.2	Solution of simple trigonometric equations in a given interval.	Students should be able to solve equations such as $\sin\left(x + \frac{\pi}{2}\right) = \frac{3}{4} \quad \text{for } 0 < x < 2\pi,$ $\cos(x + 30^\circ) = \frac{1}{2} \quad \text{for } -180^\circ < x < 180^\circ,$ $\tan 2x = 1 \quad \text{for } 90^\circ < x < 270^\circ,$ $6 \cos^2 x + \sin x - 5 = 0 \quad \text{for } 0 \leq x < 360^\circ,$ $\sin^2\left(x + \frac{\pi}{6}\right) = \frac{1}{2} \quad \text{for } -\pi \leq x < \pi.$